

智能训练_机器学习回归问题训练模型示例

```
1  ✓ # 先安装需要的库
2  # pip install numpy pandas scikit-learn -i https://pypi.tuna.tsinghua.edu.cn/simple
3  # 这是一个机器学习回归问题，
4  ✓ import pandas as pd
5  import numpy as np
6  from sklearn.model_selection import train_test_split
7  from sklearn.metrics import mean_squared_error
8  from sklearn.neighbors import KNeighborsRegressor
9  from sklearn.preprocessing import StandardScaler
10 # 加载数据
11 data = pd.read_csv('data2.csv')
12 # 处理缺失值（用均值填充）
13 data.fillna(data.mean(), inplace=True)
14 # 删除完全重复的行
15 data.drop_duplicates(inplace=True)
16 # 特征和目标变量
17 X = data[['人均收入', '家庭平均收入']]
18 y = data['当地平均房价']
19 # 标准化特征
20 scaler = StandardScaler()
21 X_scaled = scaler.fit_transform(X)
22 # 划分训练集和测试集
23 X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=2020)
24 # K近邻回归
25 model = KNeighborsRegressor(n_neighbors=5)
26 model.fit(X_train, y_train)
27 y_pred = model.predict(X_test)
28 mse = mean_squared_error(y_test, y_pred)
29 print(f'均方误差 (MSE): {mse}')
30 # 计算 RMSE
31 rmse = np.sqrt(mse)
32 print(f'均方根误差 (RMSE): {rmse}')
33 print(model.score(X_test, y_test))
```

智能训练_机器学习分类问题训练模型示例

```
1 # 先安装需要的库
2 # pip install numpy pandas scikit-learn -i https://pypi.tuna.tsinghua.edu.cn/simple
3
4 # 这是一个机器学习分类问题
5 import pandas as pd # 导入pandas库
6 from sklearn.model_selection import train_test_split # 导入机器学习库scikit-learn
7 from sklearn.linear_model import LogisticRegression # 导入逻辑回归模型
8
9 data = pd.read_csv("data.csv") # 读取data.csv数据文件
10 print(data.head(15)) # 显示数据前15行
11 print(data.tail(15)) # 显示数据后15行
12 print(data.describe()) # 显示数据描述统计信息
13 data = data.dropna() # 去除缺失值
14 data = data.drop_duplicates() # 去除重复值
15
16 X = data["预估薪资"] # 提取特征列
17 y = data['是否会购买'] # 提取标签
18
19 # 按8:2的比例划分训练集和测试集
20 X_train, X_test, y_train, y_test = train_test_split(*arrays: X, y, test_size=0.2, random_state=2025)
21 X_train = X_train.values.reshape(-1, 1) # 将训练集特征转换为二维数组ndarray
22 X_test = X_test.values.reshape(-1, 1) # 将测试集特征转换为二维数组ndarray
23
24 model = LogisticRegression() # 初始化逻辑回归模型
25 model.fit(X_train, y_train) # 开始训练
26 print(model.score(X_test, y_test)) # 打印评价分数
27 print(model.predict(X_test)) # 打印预测结果
```

常用算法模型库

```
34 # 以下是常用的一些算法，有精力可以看一看，全部背下来不太容易
35 from sklearn.neural_network import MLPRegressor
36 from sklearn.linear_model import LinearRegression
37 from sklearn.tree import DecisionTreeRegressor
38 from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
39 from sklearn.svm import SVR
40 # 定义模型
41 models = {
42     '线性回归': LinearRegression(),
43     '决策树回归': DecisionTreeRegressor(random_state=2020),
44     '随机森林回归': RandomForestRegressor(n_estimators=100, random_state=2020),
45     '梯度提升回归': GradientBoostingRegressor(n_estimators=100, random_state=2020),
46     '支持向量回归': SVR(kernel='rbf'),
47     'K近邻回归': KNeighborsRegressor(n_neighbors=5),
48     '神经网络回归': MLPRegressor(hidden_layer_sizes=(100, 50), max_iter=1000, random_state=2020)
49 }
50 # 训练和评估模型
51 for name, model in models.items():
52     model.fit(X_train, y_train)
53     y_pred = model.predict(X_test)
54     mse = mean_squared_error(y_test, y_pred)
55     print(f'{name} 的均方误差 (MSE): {mse}')
56     # 计算 RMSE
57     rmse = np.sqrt(mse)
58     print(f'{name} 的均方根误差 (RMSE): {rmse}')
```